



Welcome to CSC 365

Internet Programming

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HTML Background Images

To add a background image on an HTML element, use the HTML style attribute and the CSS background-image property:

- `<body style="background-image: url('./imgs/MSU_bear_head.png');">`

To avoid the background image from repeating itself, set the `background-repeat` property to `no-repeat`.

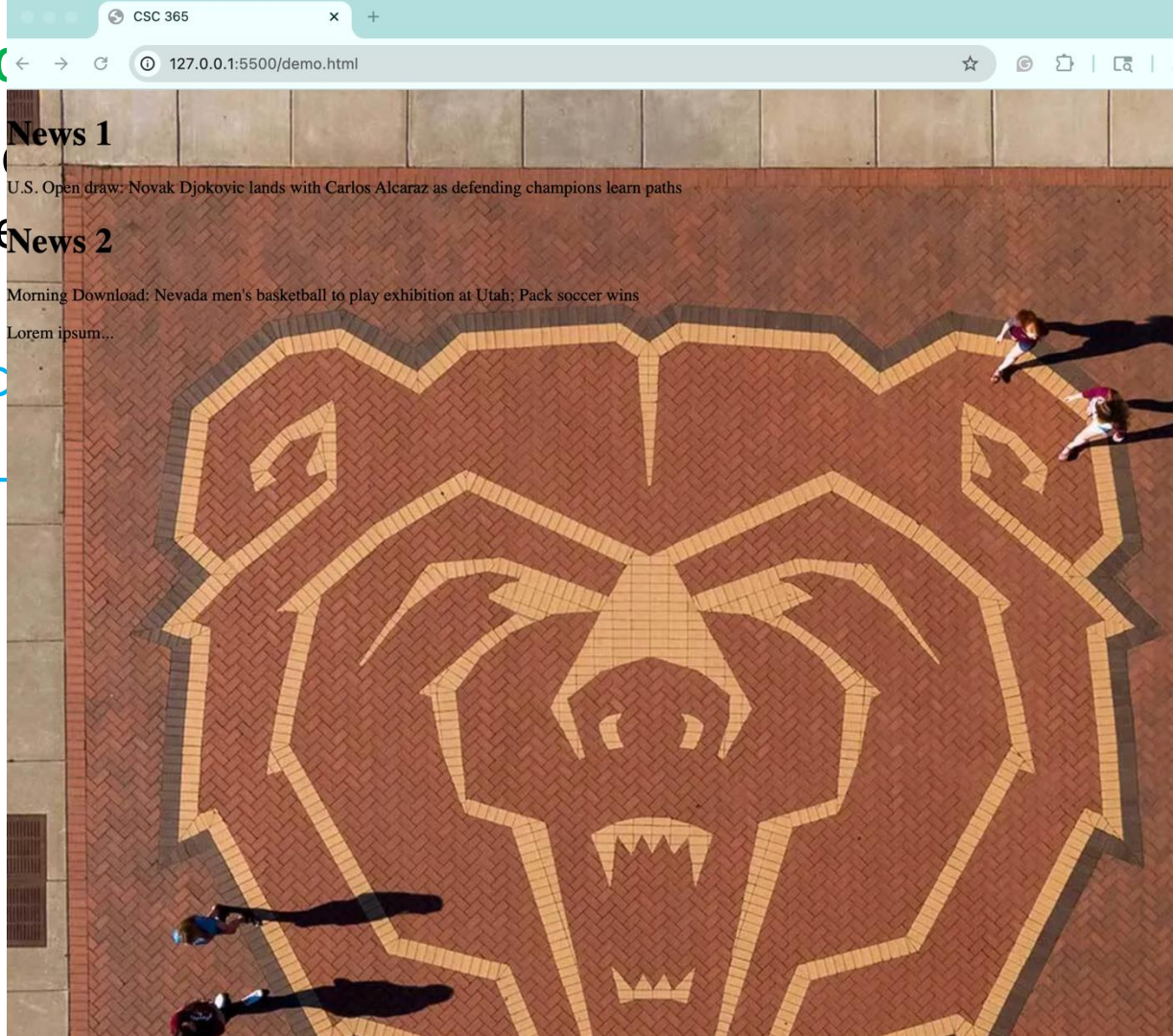
HTML Background

To add a background

HTML style attribute

property:

- `<body style="background-image: url('./imgs/MSU_`



HTML Favicon

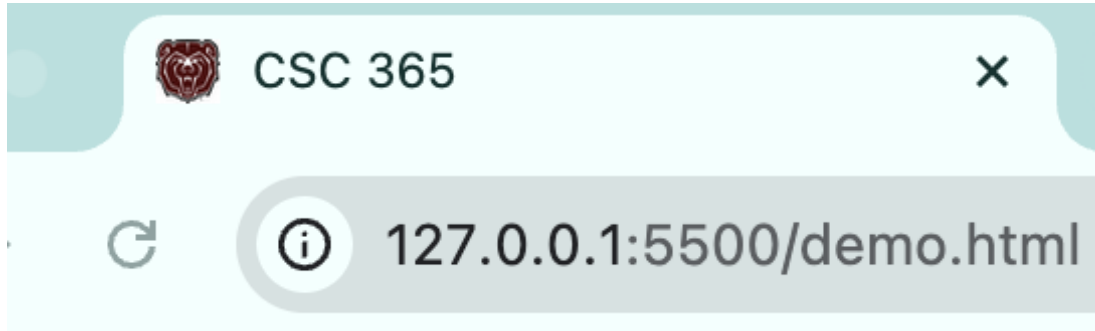
Add a <link> element to the <head> tag in html file, after the <title> element.

- `<link rel="icon" type="image/bearhead" href="imgs/bearhead.jpg">`

HTML Favicon

Add a <link> element to the <head> tag in html file, after the <title> element.

- `<link rel="icon" type="image/bearhead" href="imgs/bearhead.jpg">`



HTML Block and Inline Elements

1. Block-Level Elements.

- Block elements take up the full width available, starting on a new line.

2. Inline Elements

- Inline elements only take up as much width as necessary.

Feature	Block Elements	Inline Elements
Starts on new line?	Yes	No
Width	Full available width	As wide as the content
Can contain	Block + Inline elements	Only Inline elements & text
Examples	<code><div></code> , <code><p></code> , <code><h1></code> – <code><h6></code> , <code></code> , <code></code>	<code></code> , <code><a></code> , <code></code> , <code></code>

HTML class Attribute

- The HTML class attribute is used to specify a class for an HTML element.
- The HTML class attribute is one of the most important attributes in web development because it allows you to group and style elements, as well as manipulate them with JavaScript.

HTML class Attribute

What It Does?

The class attribute assigns one or more class names to an HTML element.

These class names can be targeted by CSS (for styling) or JavaScript (for behavior).

To create a class; write a period (.) character, followed by a class name. Then, define the CSS properties within curly braces {}:


```

1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <!-- <link rel="stylesheet" href="styles.css" -->
6      <style>
7          .city {
8              background-color: tomato;
9              color: white;
10             border: 2px solid black;
11             margin: 20px;
12             padding: 20px;
13         }
14
15         .note {
16             font-size: 120%;
17             color: blue;
18         }
19     </style>
20     <meta charset="UTF-8">
21     <meta name="viewport" content="width=device-width, initial-scale=1.0">
22     <title>CSC 365</title>
23     <link rel="icon" type="image/bearhead" href="imgs/bearhead.jpg">
24 </head>
25
26 <body>
27     <div class="city main">
28         <h2>Kansas City</h2>
29         <p>Kansas City, Missouri, <span class="note">abbreviated KC</span> or KCMO, is the largest city in the state of
30             Missouri by population and
31             area.</p>
32     </div>
33
34     <div class="city">
35         <h2>St. Louis</h2>
36         <p>It is the largest metropolitan area in Missouri and the second-largest in Illinois.</p>
37     </div>
38
39     <div class="city">
40         <h2>Springfield</h2>
41         <p>Springfield is the <span class="note">third</span> most populous city in the state of Missouri and the county
42             seat of Greene County.
43         </p>
44     </div>
45 </body>
46
47 </html>

```

```

demo.html > html > body
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5    <!-- <link rel="stylesheet" href="style
6    <style>
7      .city {
8        background-color: tomato;
9        color: white;
10       border: 2px solid black;
11       margin: 20px;
12       padding: 20px;
13     }
14
15     .note {
16       font-size: 120%;
17       color: blue;
18     }
19   </style>
20   <meta charset="UTF-8">
21   <meta name="viewport" content="width=de
22   <title>CSC 365</title>
23   <link rel="icon" type="image/bearhead"
24 </head>
25
26 <body>
27   <div class="city main">
28     <h2>Kansas City</h2>
29     <p>Kansas City, Missouri, <span clas
30       Missouri by population and
31       area.</p>
32   </div>
33
34   <div class="city">
35     <h2>St. Louis</h2>
36     <p>It is the largest metropolitan a
37   </div>
38
39   <div class="city">
40     <h2>Springfield</h2>
41     <p>Springfield is the <span class="
42       seat of Greene County.
43     </p>
44   </div>
45 </body>
46
47 </html>

```



Kansas City

Kansas City, Missouri, **abbreviated KC** or KCMO, is the largest city in the state of Missouri by population and area.

St. Louis

It is the largest metropolitan area in Missouri and the second-largest in Illinois.

Springfield

Springfield is the **third** most populous city in the state of Missouri and the county seat of Greene County.

HTML class Attribute

HTML elements can belong to more than one class. To define multiple classes, separate the class names with a space, e.g. `<div class="city main">`.

An element can have more than one class, separated by spaces. This allows you to combine styles.

Class name can be utilized by JavaScript, we will look into it later.

HTML id Attribute

The HTML id attribute is used to uniquely identify a single element on a webpage. It plays an important role in styling, scripting, and navigation.

What It Does?

- Assigns a unique identifier to an element.
- Each id must be unique within a page (unlike class, which can be reused).

Styling with id: You can target an element by its id using the hash symbol (#) in styling code.

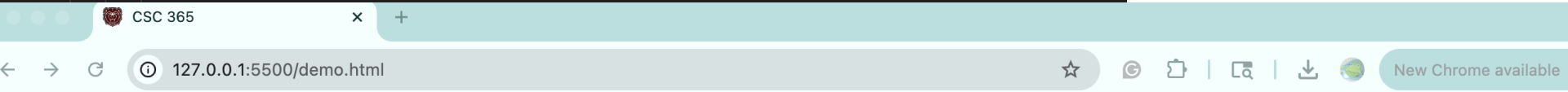
id name can be utilized by JavaScript, we will look into it later.

```

1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <!-- <link rel="stylesheet" href="styles.css"> -->
6      <style>
7          .city {
8              background-color: #tomato;
9              color: #white;
10             border: 2px solid #black;
11             margin: 20px;
12             padding: 20px;
13         }
14
15         #myHeader {
16             background-color: #lightblue;
17             color: #black;
18             padding: 40px;
19             text-align: center;
20         }
21     </style>
22     <meta charset="UTF-8">
23     <meta name="viewport" content="width=device-width, initial-scale=1.0">
24     <title>CSC 365</title>
25     <link rel="icon" type="image/bearhead" href="imgs/bearhead.jpg">
26 </head>
27
28 <body>
29     <div class="city main">
30         <h2 id="myHeader">Kansas City</h2>
31         <p>Kansas City, Missouri, <span class="note">abbreviated KC</span> or KCMO, is the largest city in the state of
32             Missouri by population and
33             area.</p>
34     </div>
35
36 </body>
37
38 </html>

```

```
demo.html > html > body
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5    <!-- <link rel="stylesheet" href="styles.css"> -->
6    <style>
7      .city {
8        background-color: tomato;
9        color: white;
10       border: 2px solid black;
```



Kansas City

Kansas City, Missouri, abbreviated KC or KCMO, is the largest city in the state of Missouri by population and area.

```
29  <div class="city main">
30    <h2 id="myHeader">Kansas City</h2>
31    <p>Kansas City, Missouri, <span class="note">abbreviated KC</span> or KCMO, is the largest city in the state of
32      Missouri by population and
33      area.</p>
34  </div>
35
36 </body>
37
38 </html>
```

HTML id Attribute

What else it Does?

Linking to an Element (Anchor Navigation)

Webpage could be long, HTML **id** can be used to allow readers to jump to specific parts of a webpage.

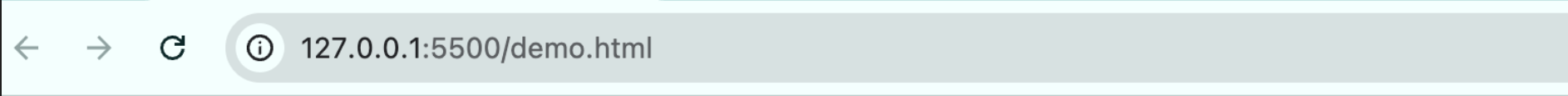
```

demo.html > html
1  <!DOCTYPE ht
2  <html lang="en"
3
4  <head>
5      <!-- <link rel="stylesheet" href="styles.css" -->
6
7      <meta charset="UTF-8">
8      <meta name="viewport" content="width=device-width, initial-scale=1.0">
9      <title>CSC 365</title>
10     <link rel="icon" type="image/bearhead" href="imgs/bearhead.jpg">
11 </head>
12
13 <body>
14     <a href="#contact">Go to Contact Section</a>
15
16     <h2>Kansas City</h2>
17     <p>Kansas City, Missouri, <span class="note">abbreviated KC</span> or KCMO, is the largest city in the state of
18         Missouri by population and
19         area.</p>
20
21     <h2 id="contact">Contact Us</h2>
22     <p>Here's our contact info...</p>
23 </body>
24
25 </html>

```



```
<> demo.html > html
1 <!DOCTYPE ht html (field)
2 <html lang="en">
```



[Go to Contact Section](#)

Kansas City

Kansas City, Missouri, abbreviated KC or KCMO, is the largest city in the state of Missouri by population and area.

Contact Us

Here's our contact info...

```
25 </html>
```

Difference Between id and class

Attribute	Usage	Uniqueness
id	Identifies a single element (used for unique styles, scripting, or anchors).	Must be unique in a page
class	Groups multiple elements with shared styles/behavior.	Can be reused across many elements

CSS: Cascading Style Sheets

CSS is a stylesheet language used to control the presentation and styling of HTML documents.

Why is CSS Important?

1. Separation of Concerns

- HTML handles structure.
- CSS handles style.
- This makes websites easier to maintain and update.

CSS: Cascading Style Sheets

Why is CSS Important?

2. Consistency

- A single CSS file can control the style of an entire website.
- If you change a color or font in the CSS, it updates everywhere.

2. Flexibility

- Enables responsive design (websites that adapt to mobile, tablet, and desktop).
- Allows animations, transitions, and modern design features.

CSS:

How CSS is Added?

- Inline CSS (inside an element).
- Internal CSS (inside <style> in the HTML <head>).
- External CSS (linked file, recommended):

Simply create a file named styles.css

Utilize the following code to link css file and html file

```
<link rel="stylesheet" href="styles.css">
```

```
<> demo.html >  html
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <link rel="stylesheet" href="styles.css">
6      <meta charset="UTF-8">
7      <meta name="viewport" content="width=device-width, initial-scale=1.0">
8      <title>CSC 365</title>
9      <link rel="icon" type="image/bearhead" href="imgs/bearhead.jpg">
10 </head>
11
12 <body>
13     <h1>News 1</h1>
14     <p>U.S. Open draw: Novak Djokovic lands with Carlos Alcaraz as defending champions learn paths</p>
15
16     <h2>News 2</h2>
17     <p>Morning Download: Nevada men's basketball to play exhibition at Utah; Pack soccer wins</p>
18     <p>Lorem ipsum...</p>
19 </body>
20
21 </html>
```

```
# styles.css > p
1  body {
2    background-image: url('./imgs/MSU_bear_head.png');
3  }
4  h1 {
5    color: blue;
6    font-family: verdana;
7    font-size: 300%
8  }
9  p {
10   color: green;
11   font-family: courier;
12   font-size: 160%;
13   border: 2px solid powderblue;
14 }
```

```
# styles.css > p
1   body {
2   background-image: url('/imgs/MSU_bear_head.png');
```

CSC 365

127.0.0.1:5500/demo.html

New Chrome available

News 1

U.S. Open draw: Novak Djokovic lands with Carlos Alcaraz as defending champions learn paths

News 2

Morning Download: Nevada men's basketball to play exhibition at Utah; Pack soccer wins

Lorem ipsum...

In class activity 3

Create a CSS file and write your CSS rules in it to style the HTML content.

Next Class



Week 2:

Topic:

- HTML and CSS